

ASearcher-8B: key training setups & profiling status

Agentic-RL-CA rung-3 — branch `sync-partial-rollout-poc`

2026-07-30 04:53 CST (snapshot)

1. Key setups (current values; where set)

Setting	Value	Notes / rationale
Model / data	Qwen3-8B-Base; Base-35k unfiltered; wiki-18/e5 top-5	RL-from-base, no cold start (ASearcher recipe); seed 0
Horizon / context	32 turns nominal; <code>MAX_PROMPT=16384</code> ; <code>MAX_RESPONSE=1024/turn</code>	16k binds first: measured mean ~ 6 turns, max 26; 24–37% of turns hit the 1024 cap
History	full: past <code><think>+<search>+ 5k-char <information></code> blocks, never dropped	<code>env.asearcher_history</code> ; 5k cap hits 3–8% of retrievals (<code>info_char_cap=5000</code>)
Truncation	<code>clean</code> <code>ctx-terminate</code> (<code>env.ctx_terminate</code> , margin 256); <code>TRUNCATION=left</code> as never-fire valve	episode ends before the model sees a truncated prompt; residual clip 0.5% (prefilter tail — tighten before arms)
Partial rollout	PoC, per-run flag; <code>cycle_turns=8</code> , <code>max_age=4</code> , group-atomic release, per-row <code>policy_version</code> ; no off-policy correction	T1 resume identity byte-exact; invariants exact; 0 stale drops; $1.5\text{--}1.6\times$ measured ($2.2\text{--}2.5\times$ gen)
Prefix caching (vLLM)	ON but defeated by template (step-count precedes history)	only ~ 200 of $\sim 4.6\text{k}$ mean prompt tokens reusable; template fix = new MDP, deferred to a protocol boundary
Chunked prefill	ON as of <code>grpo_fast</code> + <code>max_num_batched_tokens=17408</code>	fork requires $\geq \text{max_model_len}$ (user-requested 16384 rejected by validation — deviation)
Dynamic batching	<code>DYNBSZ=1: use_dynamic_bsz @ 24,576 tok/GPU</code> (actor/logprob/ref; critic keys via cond) — under test now	replaces fixed <code>MICRO_BSZ=1</code> ($\sim 5.2\text{k}$ real tok/forward); gradient-identical; OOM ladder $24576 \rightarrow 16384 \rightarrow \text{revert}$
Batching (fixed)	$256 \times 5 = 1280$ traj/step; mini 512 rows; <code>remove_padding</code> , <code>balance_batch</code> , grad-ckpt all ON	matched to 4B arms (budget comparability)
Memory posture	KV frac 0.5; critic param+optim offload (gae arms)	actor-only peaked 85GB at KV 0.6; critic OOM'd at <code>wake_up(kv_cache)</code> without offload
Algorithm (fixed)	GRPO vs turnPPO; $\gamma=\lambda=1$; token-mean loss; clip 0.2 (dual 3.0); GRPO KL $1\text{e-}3$ / turnPPO no KL; entropy 0	§4 of the config audit; unchanged by any speed work

2. Profiling matrix status (04:53 CST)

Run	Status	Key result / ETA
p8bprof_grpo_vanilla	done	7,652 s/step; 5.98 s/traj; 13.3 d/arm
p8bprof_grpo_partial	done	4 cycles; 4.00 s/traj; 8.9 d/arm
p8bprof_ppo_partial	done	3 cycles; 5.76 s/traj; 12.8 d/arm
p8bprof_ppo_vanilla (retry)	running	step 1 in progress under KV 0.5+offload; ETA cfg done $\sim 09\text{--}11$ CST
p8bprof_grpo_fast	running	DYNBSZ 24576 + chunked prefill + partial; attempt live since 04:27, step 1 $\sim 05\text{:}10$, preli

Also in flight: 4B p3v2 — token-GRPO **done** (final val@150 **em 0.773**, 3.4 turns); turn-PPO **running** step 73/150 (0.447@50; val@75 next; ETA finish mid-day Jul 30). 8B arm launch: **held** pending **grpo_fast** numbers + your choice (partial/steps/vanilla). Quota 6/6: p3v2-turn-ppo, rung-4×2 (parallel session), ppo-retry2, grpo-fast3.

3. Cost history of the fast-config attempts (transparency)

Attempt 1: `set -u` unbound `ADV_ESTIMATOR` in protocol DYNBSZ block (fixed; critic keys moved to `run_condition`).
Attempt 2: fork validation `max_num_batched_tokens ≥ max_model_len` (fixed: 17408 derived). Attempt 3: **running** clean past both failure points. Total overhead ~50 min.

4. Pointers

Config audit: `docs/design/2026-07-30_long_horizon_config_audit.md`. Partial-rollout design: `docs/design/2026-07-29_sync_par`
Full profiling report: `docs/reports/2026-07-29_p8b_profiling/p8b_profiling_report.pdf`. All docs also at `~/xiaoxuan/arlca-8b/`
wandb project [rainyfields/ca-rung3-8b-asearcher](#).